

Genomic Data Security

Yongjin Park

18 March 2026

OPINIONS ON INTERNET PRIVACY

THE PHILOSOPHER:

"PRIVACY" IS AN IMPRACTICAL WAY TO THINK ABOUT DATA IN A DIGITAL WORLD SO UNLIKE THE ONE IN WHICH OUR SOCI-

SO BORED.



THE CRYPTO NUT:

MY DATA IS SAFE BEHIND SIX LAYERS OF SYMMETRIC AND PUBLIC-KEY ALGORITHMS.

WHAT DATA IS IT?

MOSTLY ME EMAILING WITH PEOPLE ABOUT CRYPTOGRAPHY.



THE CONSPIRACIST:

THESE LEAKS ARE JUST THE TIP OF THE ICEBERG. THERE'S A WAREHOUSE IN UTAH WHERE THE NSA HAS THE *ENTIRE* ICEBERG.

I DON'T KNOW HOW THEY GOT IT THERE.



THE NIHILIST:

JOKE'S ON THEM, GATHERING ALL THIS DATA ON ME AS IF ANYTHING I DO MEANS ANYTHING.



THE EXHIBITIONIST:

MMMM I SURE HOPE THE NSA ISN'T WATCHING ME BITE INTO THESE JUICY STRAWBERRIES!!

OOPS I DRIPPED SOME ON MY SHIRT! BETTER TAKE IT OFF.

GOOGLE, ARE YOU THERE?

GOOGLE, THIS LOTION FEELS SOOOO GOOD.



THE SAGE:

I DON'T KNOW OR CARE WHAT DATA *ANYONE* HAS ABOUT ME.

DATA IS IMAGINARY. THIS BURRITO IS REAL.



Today's lecture

- 1 Genetic Data Utility vs. Privacy
- 2 Membership inference
- 3 Re-identification attack
- 4 Secured protocols for genomic data analysis
- 5 Differential Privacy
- 6 Summary

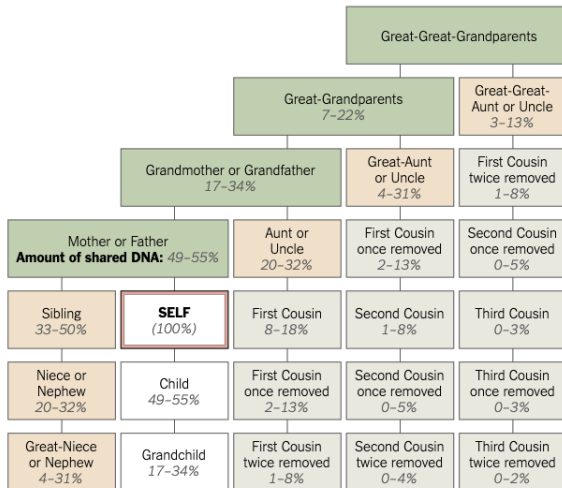


How Genetic Sleuthing Helped a Kidnapped Girl Recover Her Identity

With the help of a genomic database. . .

<https://www.nytimes.com/2018/10/15/science/gedmatch-genealogy-cold-cases.html>

Math based on (random) DNA recombination



Source: Blaine T. Bettinger

<https://www.nytimes.com/2018/10/15/science/gedmatch-genealogy-cold-cases.html>

The Golden State Killer

- **Joseph James DeAngelo** arrested in 2018 after 40+ years
- At least 13 murders, 50+ rapes across California (1970s-80s)
- First major cold case solved using Investigative Genetic Genealogy

Back story: How the case was solved



Applications

Education ▼

DNA Diaries

About ▼

Where DNA Connections Come Together

GEDmatch provides more applications for comparing your DNA test results with the most people worldwide, for free!

[VIEW OUR APPLICATIONS](#)



Back story: How the case was solved

- ① Crime scene DNA uploaded to **GEDmatch.com**

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Back story: How the case was solved

- ① Crime scene DNA uploaded to **GEDmatch.com**
- ② Found matches to **distant relatives** (not the suspect)
- ③ Reconstructed family trees back 5-8 generations
- ④ Narrowed down the pool of suspects to a handful of living individuals
- ⑤ Confirmed by collecting discarded DNA (coffee cup) from DeAngelo

How an Unlikely Family History Website Transformed Cold Case Investigations

Fifteen murder and sexual assault cases have been solved since April with a single genealogy website. This is how GEDmatch went from a casual side project to a revolutionary tool.



Sooner or Later Your Cousin's DNA Is Going to Solve a Murder

The Golden State Killer case was just the start. Hundreds of cold cases are hot again thanks to a new genealogy technique. The price may be everyone's genetic privacy.

A tale from the other side of the GSK case

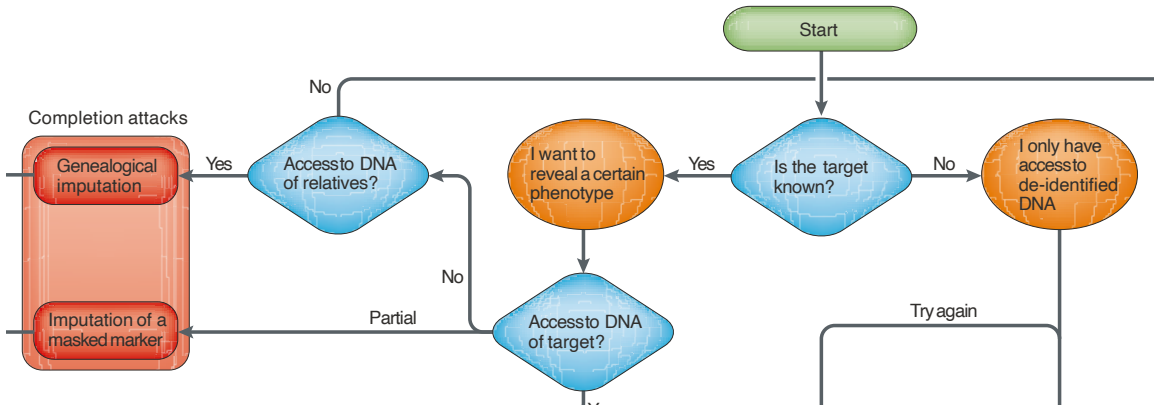
Routes for breaching and protecting genetic privacy

Yaniv Erlich¹ and Arvind Narayanan²

- 60% success with database covering just 2% of population
- As databases grow, nearly anyone can be identified
- Privacy erosion inevitable?

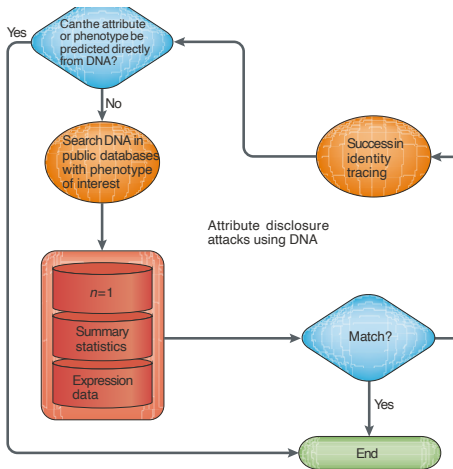
Erlich et al. (2018)

Key security breach is to find someone in DB



Erlich et al. (2018)

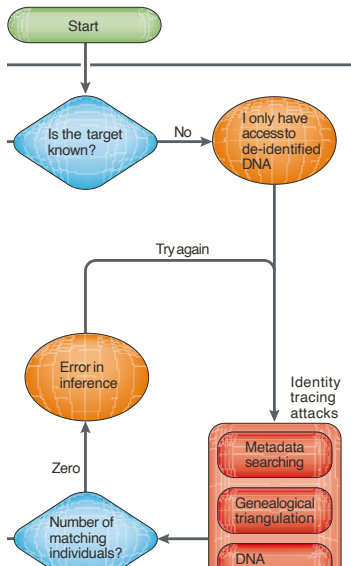
Genome-Wide Association Studies data could...



- GWAS summary statistics
- What if we can “look up” (or predict) phenotypes?
- Public genealogical records amplify genetic databases.

Erlich et al. (2018)

Narrowing down: from genealogy to individual



Yongjin Park

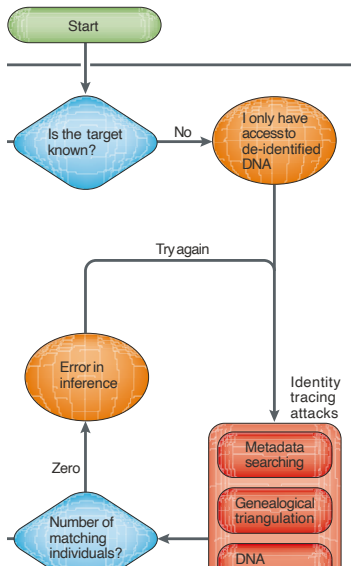
- Demographic identifiers narrow search: genealogy match → age/location → individual
- We can also narrow down to the most genomically informative locations. . .

https://en.wikipedia.org/wiki/Identity_by_descent

Genomic Data Security

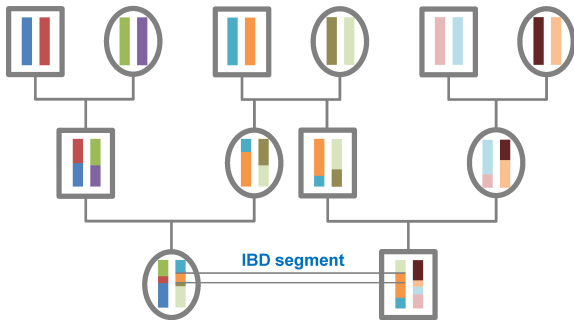
18 March 2026

Narrowing down: from genealogy to individual



Yongjin Park

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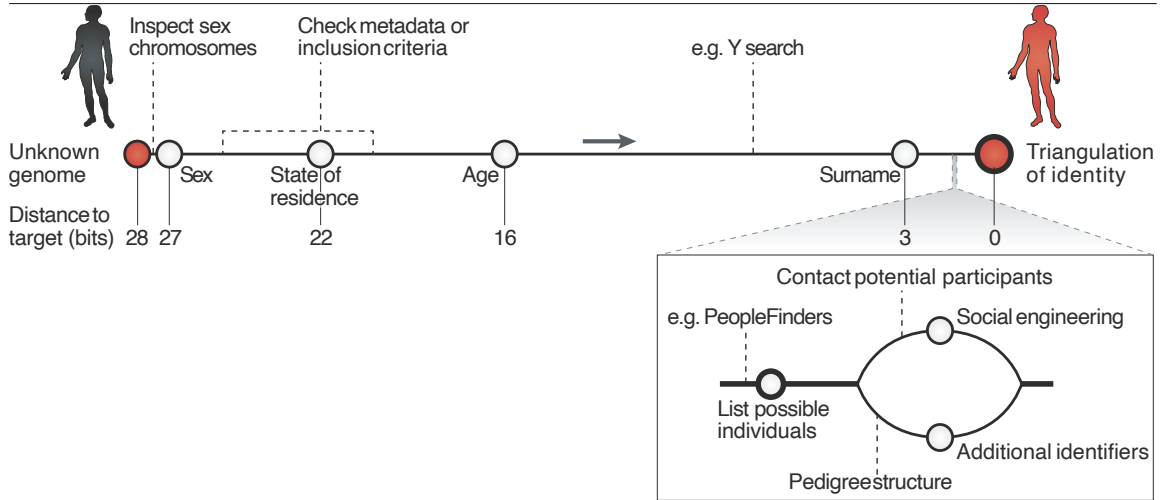


https://en.wikipedia.org/wiki/Identity_by_descent

Genomic Data Security

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Real Cases: From DNA to Identification



Erlich et al. (2018)

23andMe Files for Bankruptcy Amid Concerns About Security of Customers' Genetic Data

The company filed for Chapter 11 bankruptcy to “facilitate a sale process.” Its chief executive stepped down and said she would bid on it.

23andMe to Be Bought by Biotech Company for \$256 Million

Regeneron Pharmaceuticals said it would acquire 23andMe, which filed for bankruptcy in March and would continue to offer consumer DNA testing services.

- 15 million customers' genetic data $\approx$ \$ 256M?

23andMe Customers Did Not Expect Their DNA Data Would Be Sold, Lawsuit Claims

The genetic-testing company, which collected DNA data from users, is for sale in bankruptcy court. Now, 27 states and the District of Columbia oppose selling the data without express consent.

Privacy Concerns in the midst of corporate dynamics

- Genetic data could be sold as an asset to pay creditors
- Should genetic information have special protections in bankruptcy?
- Commercial partnerships with pharma companies for drug discovery
- Who owns data?

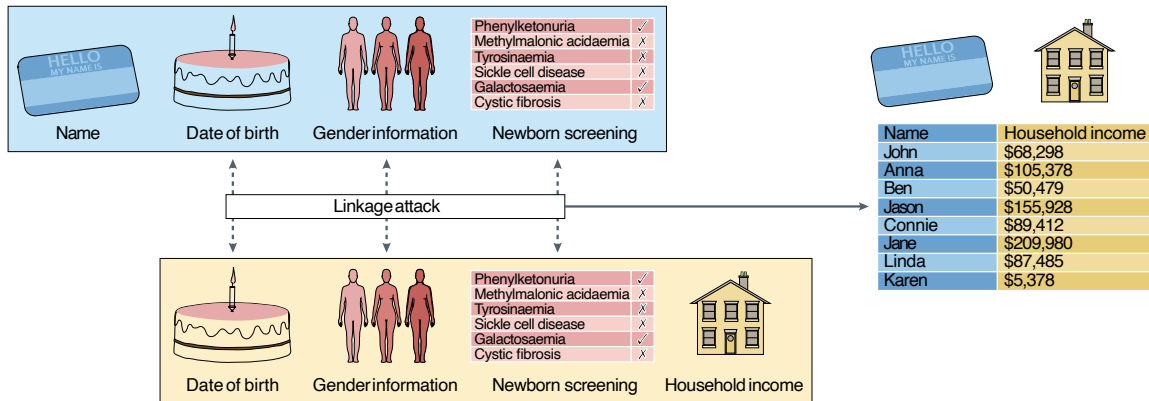
REGENERON®

Regeneron Enters into Asset Purchase Agreement to Acquire 23andMe® for \$256 Million; Plans to Maintain Consumer Genetics Business and Advance Shared Goals of Improving Human Health and Wellness

May 19, 2025 at 7:07 AM EDT

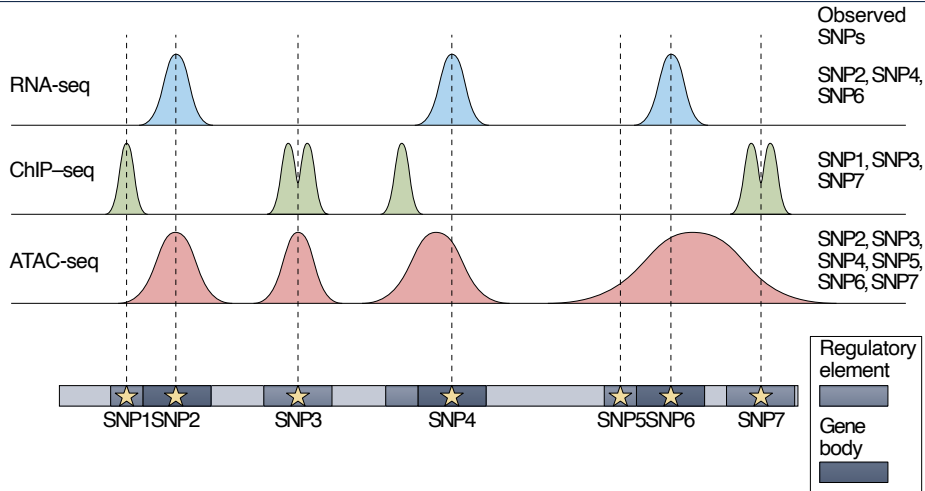
Update: On July 14, 2025, TTAM Research Institute completed the acquisition of 23andMe assets through a court-supervised bankruptcy process. Regeneron was originally declared the successful bidder in the bankruptcy auction, but once bidding was reopened, declined to submit a higher offer based on its assessment of 23andMe's remaining value.

How genetic data can compromise privacy



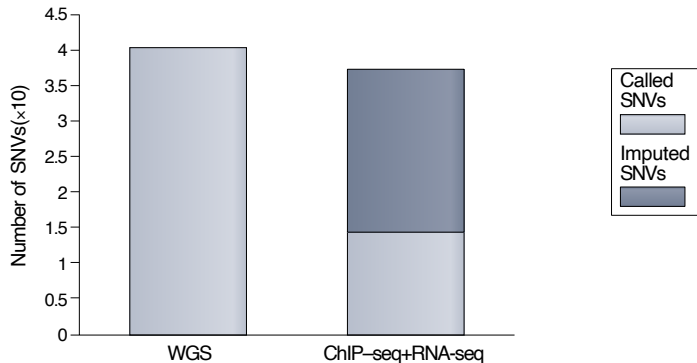
Gürsoy et al. (2022)

We may be able to infer genotypes from . . .



Gürsoy et al. (2022)

We could gain more information than we intended



Gürsoy et al. (2022)

- Single Nucleotide Variants
- Although we didn't want to know SNVs, we would be able to infer them.
- Almost every genomic technology involves "sequencing."

Knowing what the genotypes are ... could open an “unprecedented” possibility?

G → ?

GWASvariant ID	GWAScharacterizing phenotype	Hi-C		Total RNA-seq		poly(A) RNA-seq		All ChIP-seq	
		Raw	w/ imputation	Raw	w/ imputation	Raw	w/ imputation	Raw	w/ imputation
rs1667394	Blue vsbrown eye	Yes	Yes	Yes	Yes	No	No	No	Yes
rs12896399	Brown vsblond hair	Yes	Yes	No	Yes	No	No	Yes	Yes
rs1805007	Red hair	Yes	Yes	Yes	Yes	No	No	No	No
rs11648785	Freckles	No	Yes	Yes	Yes	No	Yes	No	Yes

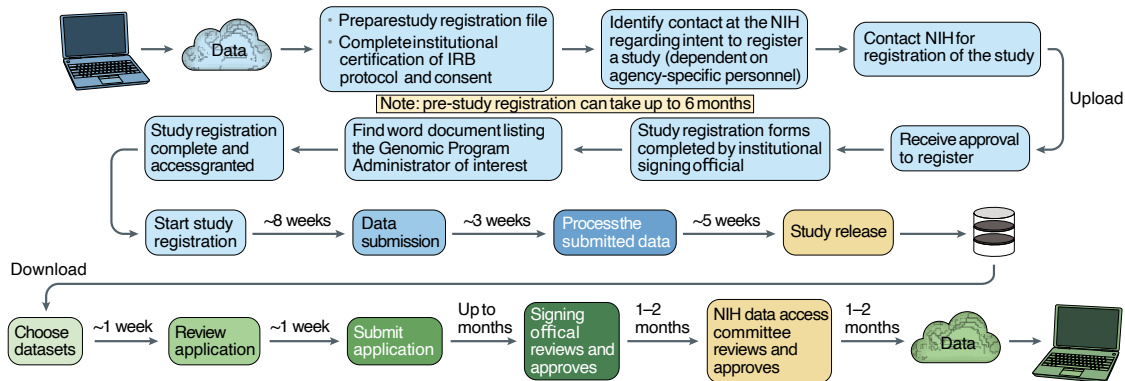
Can variant be genotyped?

Yes

No

Gürsoy et al. (2022)

After all this... are we safe?



Gürsoy et al. (2022)

Many outstanding questions...

- Should we write “no technology” manifesto like “Unabomber?”
 - What if an insurance company attempts to “predict?”
 - What if we can link them to social media?
-
- ① How genomic data security can be breached
 - ② How to prevent unwanted information leakage

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Membership attack in genomic data

Snp	Allele Frequency ($Y_{i,j}$)					Distance Measure $D(Y_{i,j}) = Y_{i,j} - \text{Pop}_j - Y_{i,j} - M_j $	Interpretation at the given SNP
	0.0	0.25	0.50	0.75	1.0		
j						$= 1.0 - 0.25 - 1.0 - 0.75 $ $= 0.75 - 0.25$ $= 0.50$	most likely to be in the Mixture
j+1						$= 0.50 - 0.250 - 0.50 - 0.75 $ $= 0.25 - 0.25$ $= 0.00$	equally likely to be in the Mixture and in the Reference Population
j+2						$= 0.00 - 0.25 - 0.00 - 0.75 $ $= 0.25 - 0.75$ $= -0.50$	most likely to be in the Reference Population

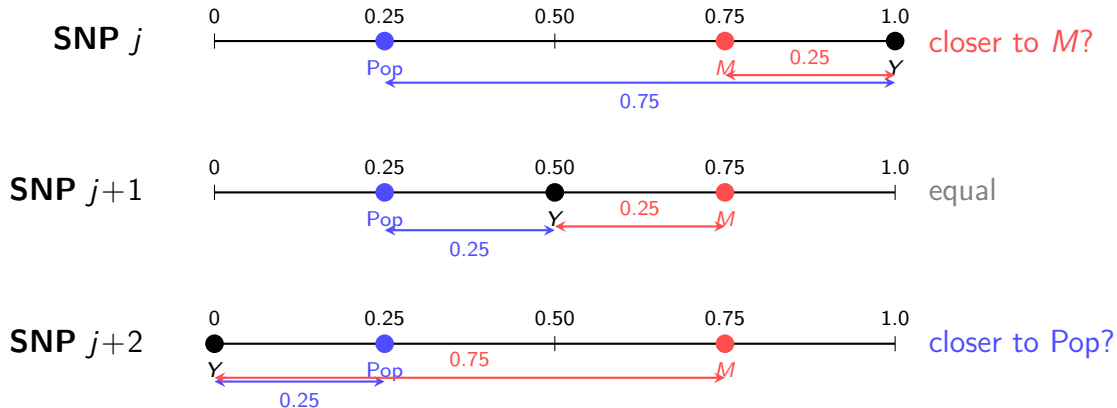
Person of Interest (Y_i)

Reference Population (Pop)

Mixture (M)

Homer et al. (2008)

Basic idea 1: Membership inference

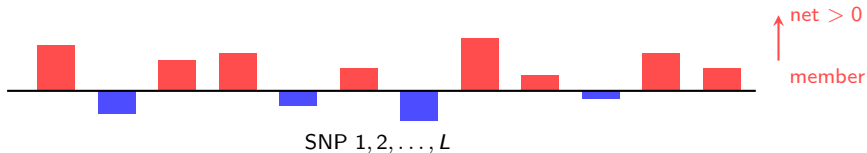


Q: Is the person of interest in the mixture pool?

Basic idea 2: Membership inference

Per SNP: $D_j = |Y_j - \text{Pop}_j| - |Y_j - M_j|$

Sum:



Test statistic:

$$T = \sum_{j=1}^L D_j \quad \begin{cases} \approx 0 & \text{non-member} \\ > 0 & \text{member} \end{cases}$$

The test statistic

Deviation of target from population at SNP j :

$$d_j = x_j - 2p_j$$

Inner product with the mixture's deviation:

$$D_{\text{mix}} = \sum_{j=1}^L \underbrace{(f_j^{\text{mix}} - p_j)}_{\text{mixture shift}} \cdot \underbrace{d_j}_{\text{target deviation}}$$

Decision rule: declare “member” if $D_{\text{mix}} > D_{\text{ref}}$

Why it works

H_0 (**not in pool**): f_j^{mix} and x_j are independent

$$E[D_{\text{mix}}] \approx 0$$

H_1 (**in pool of n**): target pulls f_j^{mix} toward x_j

$$f_j^{\text{mix}} = \frac{1}{2n} \left(x_j + \sum_{i \neq \text{target}} x_{ij} \right)$$

$$E[D_{\text{mix}} \mid H_1] \approx \frac{1}{2n} \sum_{j=1}^L d_j^2 > 0$$

Membership Test in Code

```
# Homer's membership inference test

# Target genotype:  $x$  (0, 1, or 2 for each SNP)
# Mixture allele freq:  $f_{\text{mix}}$ 
# Population allele freq:  $p_{\text{pop}}$ 

# Target's deviation from population

d <- x - 2 * p_pop

# Test statistic: inner product

D_mix <- sum((f_mix - p_pop) * d)

# Decision

is_member <- D_mix > threshold # TRUE if in pool
```

Key insight: Aggregated statistics leak membership

Signal-to-Noise Ratio

$$\text{SNR} \sim \frac{\sqrt{L}}{n}$$

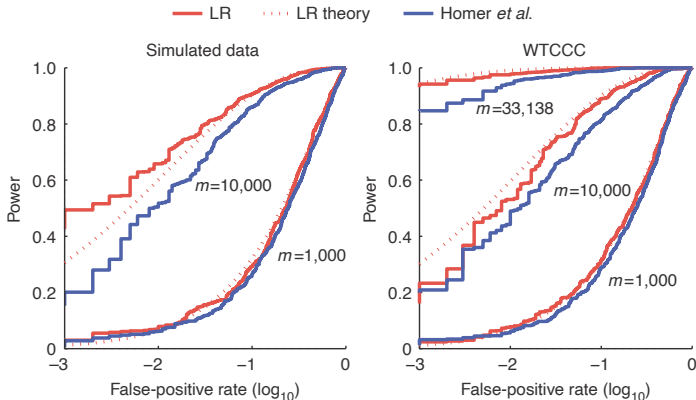
L (SNPs)	n (pool)	SNR	Detectable?
500K	100	~ 7	Yes
500K	1,000	~ 0.7	Marginal
500K	10,000	~ 0.07	No

With $L = 500\text{K}$ common SNPs, membership detectable in pools up to $n \approx 1000$ (Sankararaman et al. 2009)

Fairly strong membership prediction

Genomic privacy and limits of individual detection in a pool

Sriram Sankararaman^{1,5}, Guillaume Obozinski^{2,5},
Michael I Jordan^{1,2} & Eran Halperin^{3,4}



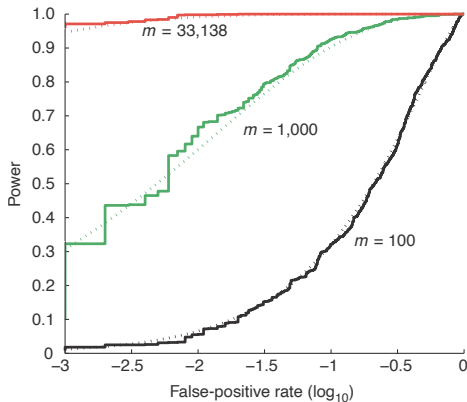
- Pool size $n = 1,000$
- LR outperforms Homer et al.
- More SNPs (m) $\Rightarrow$ higher power
- Theory $\approx$ simulation
- Validated on real WTCCC data

Fairly strong membership prediction

Result is **not population-specific** (HapMap YRI):

- Same pattern in African population
- $m = 100$: near-random
- $m = 1,000$: moderate power
- $m = 33,138$: near-perfect

Theory matches empirical across populations



HapMap YRI (African) dataset

The Beacon Attack

A **beacon** answers one query:

“Does anyone carry allele A at position X?”

Response: **yes** / **no** (1 bit)

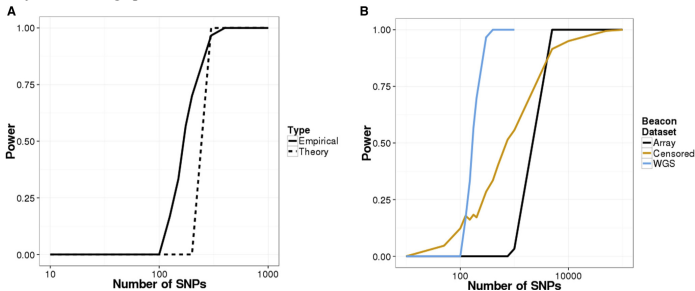
Even this minimal interface leaks membership via repeated queries
+ LR test

Shringarpure and Bustamante (2015)

ARTICLE

Privacy Risks from Genomic Data-Sharing Beacons

Suyash S. Shringarpure^{1,*} and Carlos D. Bustamante^{1,*}



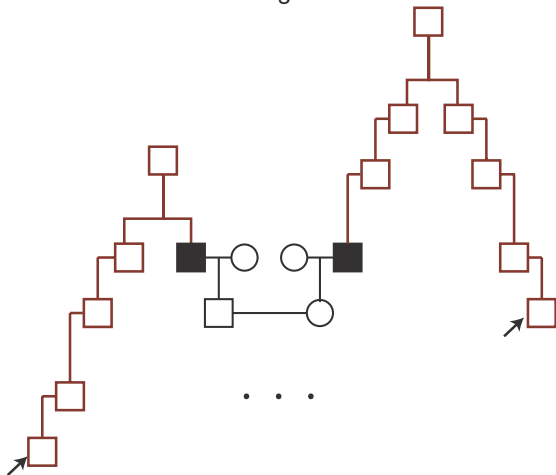
Identifying Personal Genomes by Surname Inference

Melissa Gymrek,^{1,2,3,4} Amy L. McGuire,⁵ David Golan,⁶ Eran Halperin,^{7,8,9} Yaniv Erlich^{1*}

- What's so special about Y chromosome?
- Y-STRs co-segregate with surnames; ~12% of US Caucasian males can be identified
- How does surname frequency affect vulnerability?
- Can rare surnames protect privacy?

Common surname: Y-STR reaches distant relatives

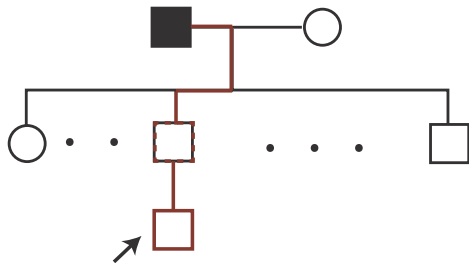
Pedigree 2



- Arrows: individuals matched via genealogy DB, 2-7 meioses from the sequenced person
- Identical Y-STR links distant relatives who never consented to data sharing
- Large pedigree $\Rightarrow$ many males share the surname, requiring additional metadata to identify

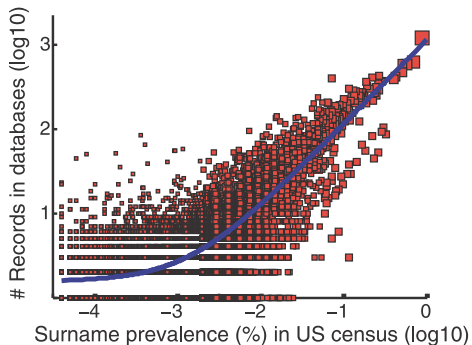
Rare surname: a single match is nearly unique

Pedigree 3



- Small lineage: few males carry this Y-STR haplotype
- Dashed: genealogy DB record holder; Arrow: identified descendant
- One match in DB $\Rightarrow$ probability of random match: 1 in 10^5 to 5×10^9
- Rare surnames are the most vulnerable to re-identification

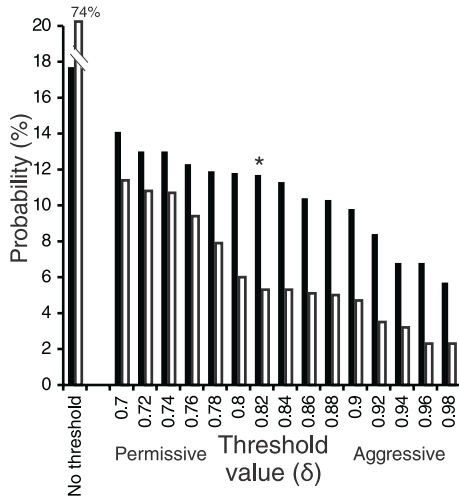
Surname coverage in genealogy databases



- Genealogy DB coverage tracks US census surname frequency
- Common surnames: many DB records $\Rightarrow$ easier to find a Y-STR match, but harder to narrow down to one person
- Rare surnames: fewer DB records, but a single match is highly identifying
- The attack surface grows as hobbyist databases expand

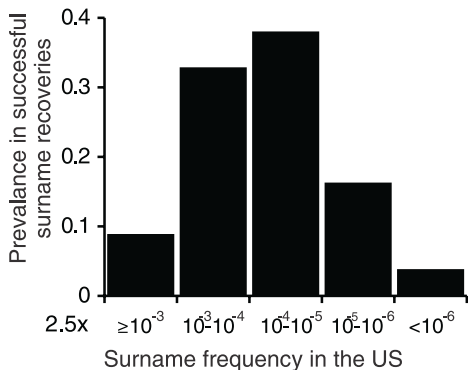
Gymrek et al. (2013)

Demographic Triangulation



- Black: surname recovery rate; Gray: false positive rate
- Trade-off: at $\delta = 0.82$, ~12% success with ~5% FPR
- Surname + age + state narrows 60,000 candidates $\rightarrow$ median of 12 individuals
- How much metadata is too much?
- What should be protected beyond HIPAA?

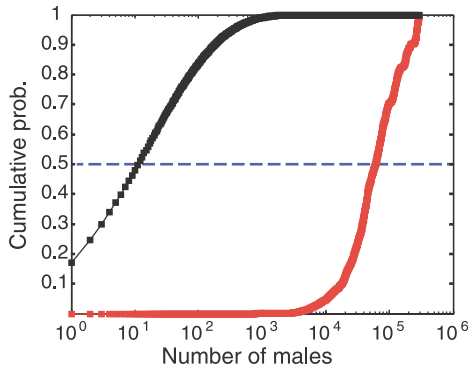
Real-World Re-identification: Craig Venter



- Wait, where is Craig Venter's DNA data?
- Clear match to "Venter" record in genetic genealogy database
- Public genomes in GenBank can be identified by name

Gymrek et al. (2013)

How much does Y-STR matching narrow the search?



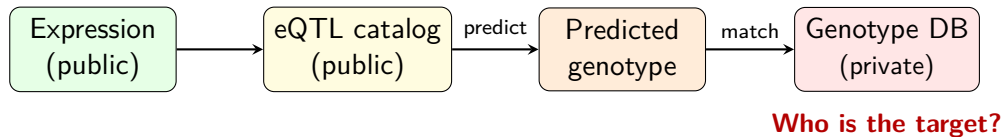
- Red: candidate pool from census alone (median $\sim 10^5$ males); Black: after Y-STR surname match (median ~ 10)
- 99% accuracy from Illumina sequencing; works even at $10\times$ coverage
- Technical noise alone does not prevent the attack

Gymrek et al. (2013)

Today's lecture

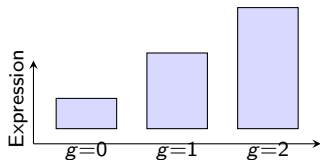
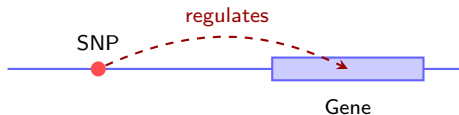
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eQTL Linking Attacks



What is eQTL?

An **expression quantitative trait locus (eQTL)** is a genomic variant (SNP) whose genotype is associated with the expression level of a gene.



$$e = \beta \cdot g + \epsilon$$

- $g \in \{0, 1, 2\}$: genotype (minor allele count)
- e : expression level
- β : eQTL effect size

Quantification of private information leakage from phenotype-genotype data: linking attacks

Arif Harmanci^{1,2} & Mark Gerstein¹⁻³

Input:

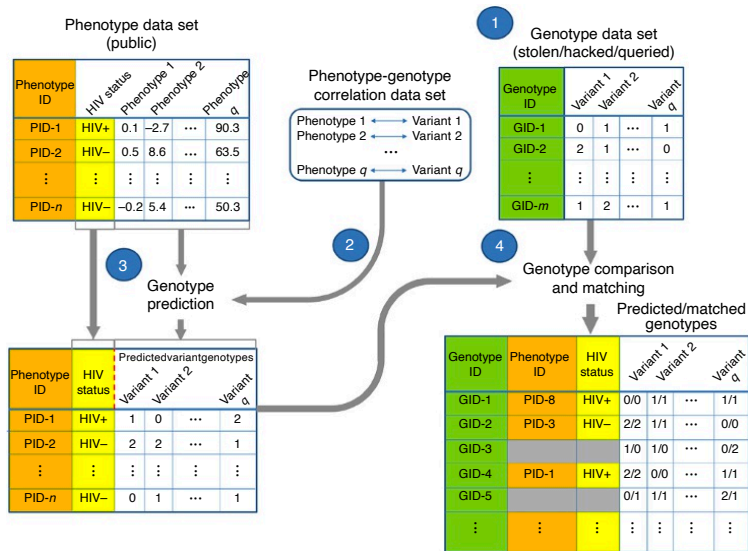
- Expression data
- eQTL summary statistics

Output:

- Genotypes

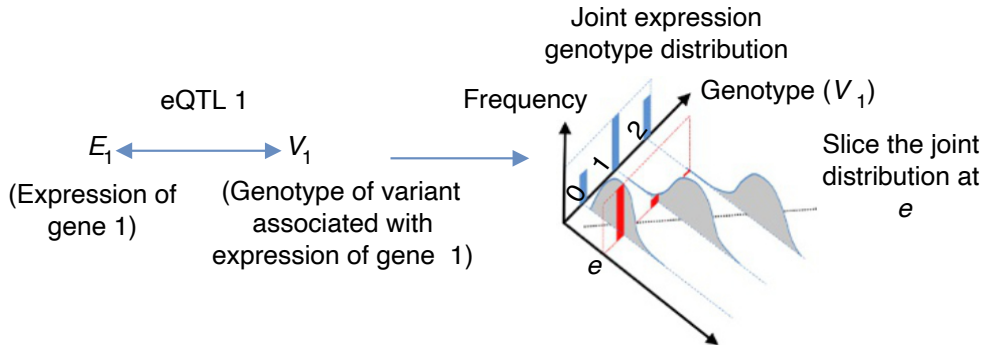
Implication: $G \rightarrow ?$

Genotypes are predictable from expression



- Correlation info links expression to variants
- Predict genotypes from observed expression
- Genotypes $\Rightarrow$ re-identification
- Would involuntarily reveal sensitive information!

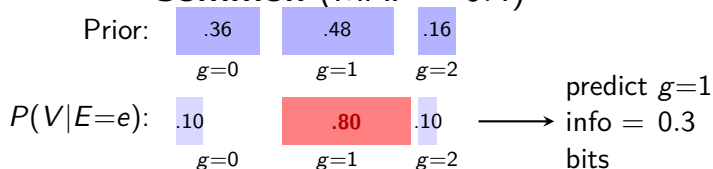
Using expression QTL as backdoor...



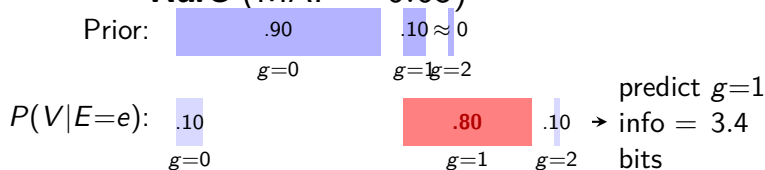
- eQTL defines a joint distribution of expression (E) and genotype (V)
- Given observed expression e , slice the joint distribution $\Rightarrow$ conditional $P(V|E = e)$
- Strong eQTL: the slice is peaked at one genotype $\Rightarrow$ high information leakage

Tradeoff: predictability vs. identifiability

Common (MAF = 0.4)



Rare (MAF = 0.05)

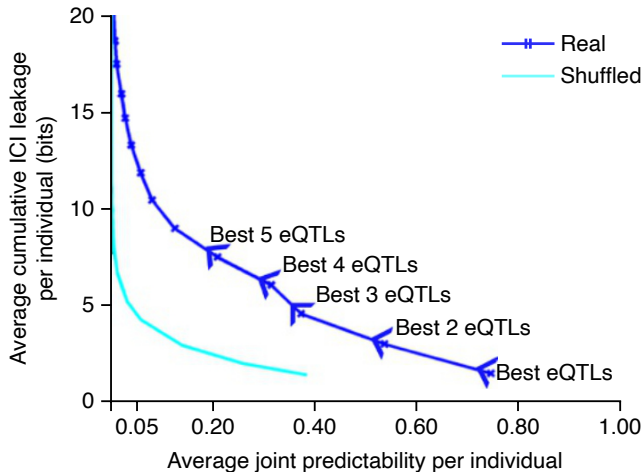


Conditional self-information:

$$\log_2 \frac{1}{P(V = g \mid E = e)}$$

- Both predict $g=1$ with 80% confidence
- Common: $g=1$ has prior .48 $\Rightarrow$ low surprise
- Rare: $g=1$ has prior .10 $\Rightarrow$ high surprise, highly identifying

Three-Step Linking Framework



- 1 Select eQTLs (correlation threshold)
 - 2 Predict genotypes from expression
 - 3 Link to database via distance metric
- Adding gender, population improves accuracy to $>95\%$
 - Even genotypes alone achieve $\sim 85\%$ accuracy

(Harmanci and Gerstein 2016)

Step 1

G-P
correlation
data set

Selection

- Absolute value of correlation

Step 2

Prediction
methodology

Genotype
prediction

- Maximum *a posteriori* genotype

Step 3

- Gender, population, age

Auxiliary
information

Linking

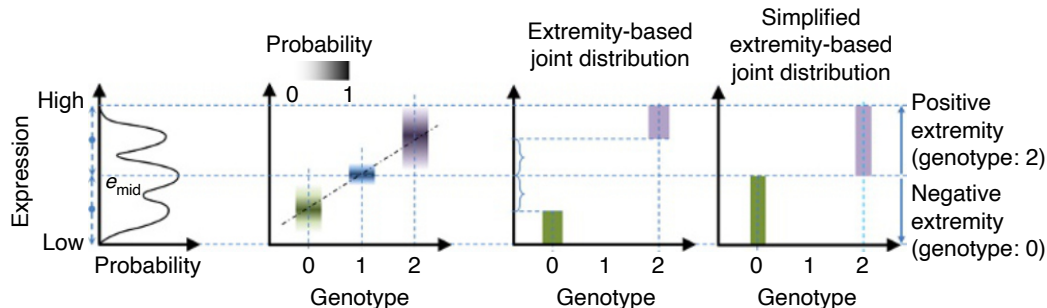
- Minimum distance between predicted and individual genotypes

Estimate
reliability of
linking

- First distance gap



Simplified attack: extremity-based prediction



- Don't need the full joint distribution: just use extreme expression values
- High expression $\Rightarrow$ predict $g = 2$; Low expression $\Rightarrow$ predict $g = 0$
- Robust across eQTL selection criteria; PPV $> 95\%$ for most thresholds

A working example on eQTL linkage attack

Attacker has (public):

- Target's expression: $\mathbf{e} = (e_1, \dots, e_K)$
- eQTL catalog: Gene $k \leftrightarrow$ SNP k , effect β_k

Attacker wants:

- Predicted genotype:
 $\hat{g}_k = \text{round}(e_k / \beta_k)$
- Match $\hat{\mathbf{g}}$ to a row in the genotype DB

Gene	Expression	β	Predicted $\hat{g}$
35	0.2	1.3	0
27	0.8	0.8	1
5	0.8	0.8	1
21	0.8	0.8	1
23	0.7	1.0	1

Match this vector against $n = 500$ individuals

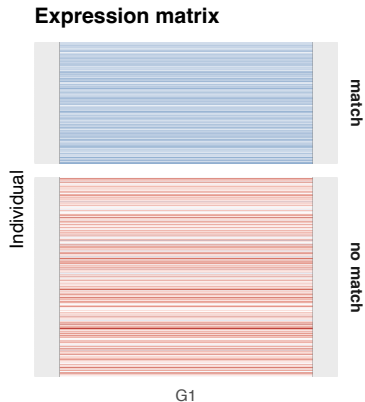
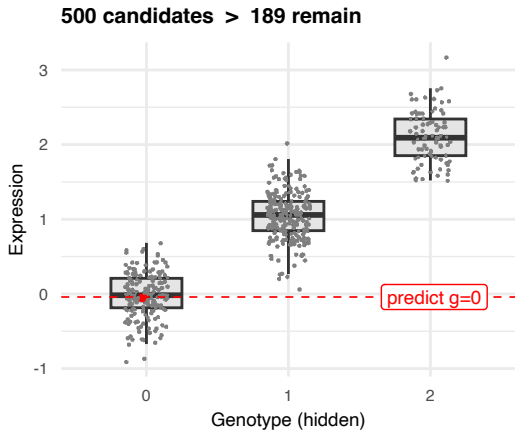
- How many unknowns?
- How many pieces of evidence?

The Attack in Code

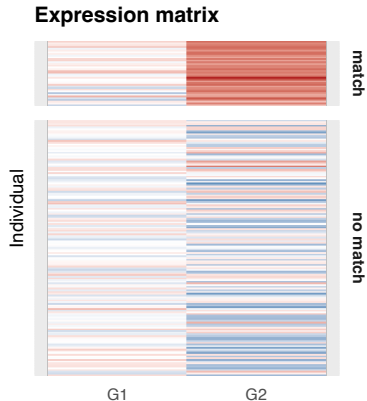
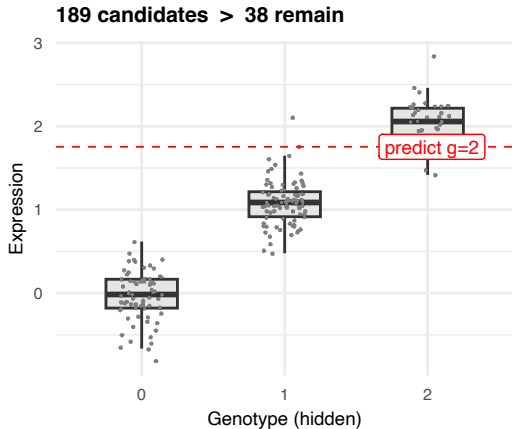
```
## eQTL model: expression = beta * genotype + noise  
E <- beta * G + rnorm(n, sd = sigma)  
  
## Predict genotype from expression (reverse the model)  
pred_genotype <- round(E / beta)  
pred_genotype <- pmin(pmax(pred_genotype, 0), 2) # clip to 0,1,2  
  
## Match predicted genotype to database  
## --> could be just nearest neighbour search  
matches <- which(G_database == pred_genotype)
```

Simple algebra can leak the identity...

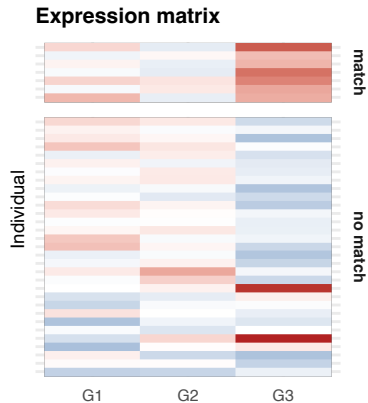
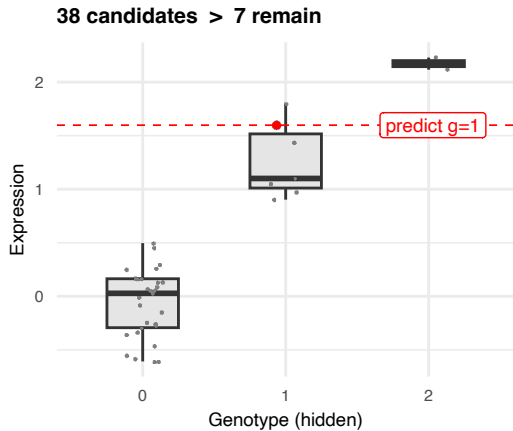
Gene 1: predicting genotype from expression



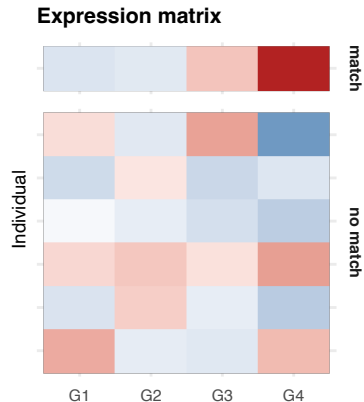
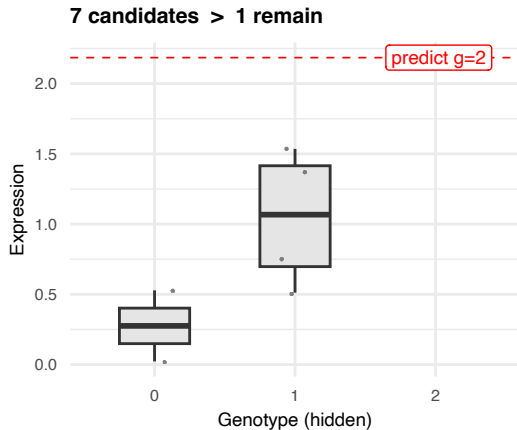
Gene 2: narrowing the pool



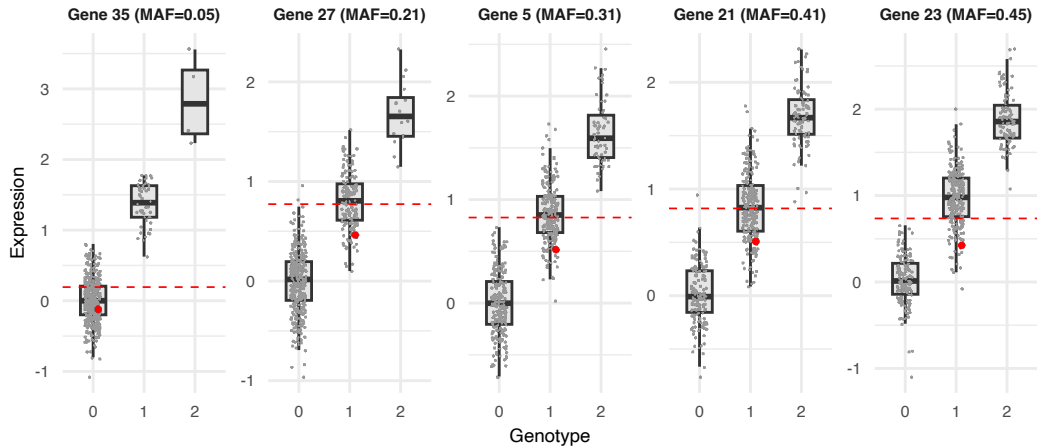
Gene 3: a rare variant



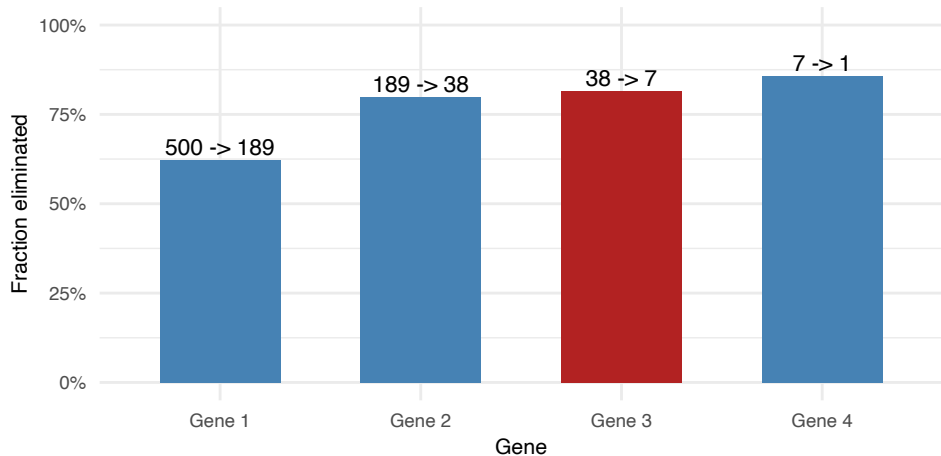
Gene 4: identified



Genotype vs. expression: the full picture



Not all genes are equal



Q: Why is a rare-variant gene more informative?

Weight matches by rarity

Exact matching is fragile: one wrong prediction breaks the chain.

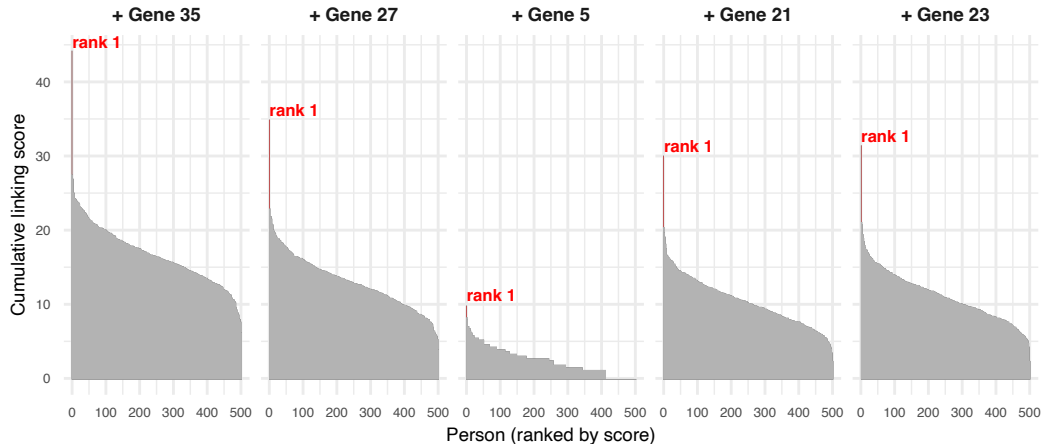
Better approach: score each match by how informative it is.

- If 45% of people have genotype $g = 1$, matching on it is **weak evidence**
- If 5% of people have genotype $g = 2$, matching on it is **strong evidence**

$$S_i = \sum_k \mathbb{I}[G_{ik} = \hat{g}_k] \cdot \underbrace{\log_2(1/f(\hat{g}_k))}_{\text{information content}}$$

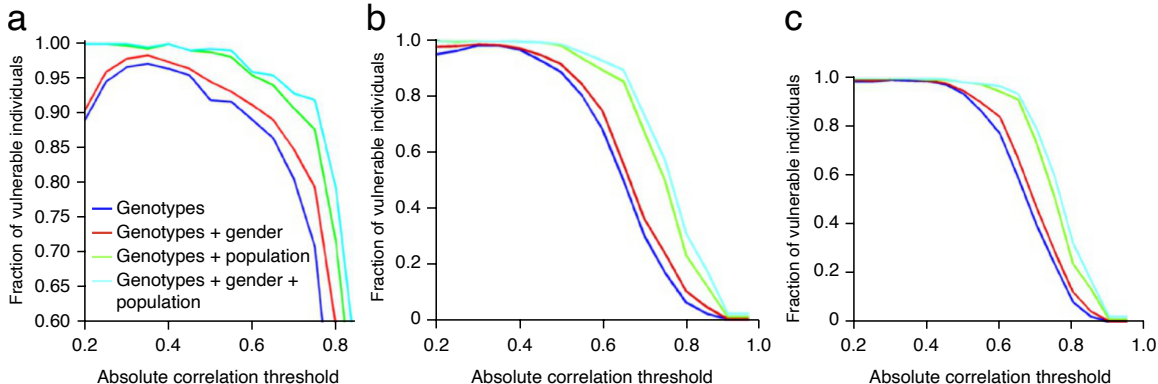
This is just information theory: rare events carry more information.

Accumulating evidence across genes



Each panel adds one gene. The target (red) moves toward rank 1 as evidence accumulates.

Auxiliary metadata amplifies the attack

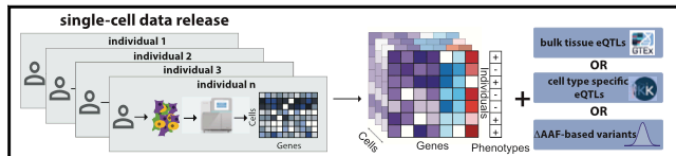


- Genotypes alone: ~85% of individuals vulnerable at moderate eQTL threshold
- Adding gender or population pushes vulnerability above 95%

(Harmanci and Gerstein 2016)

Private information leakage from single-cell count matrices

Graphical abstract

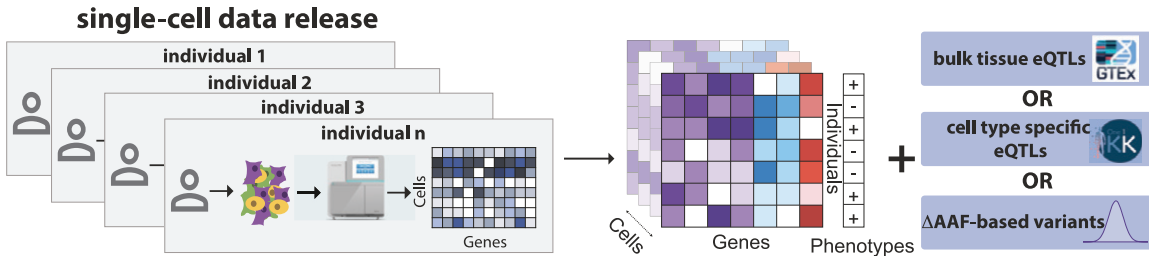


Authors

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Yoolim A. Choi, Ritambhara Singh,
Gamze Gürsoy

- Linking attack workflow: scRNA-seq → genotype prediction → database matching
- How can we share valuable single-cell data while protecting donor identity?

Linking with Bulk eQTLs



- 93-98% linking accuracy using publicly available tissue eQTLs
- Even “common” cell types leak identity. What counts as rare enough to be safe?

Walker et al. (2024)



genome database

IDs	genotypes					
	v_1	v_2	v_3		v_i	v_m
	2	2	0		1	0
	2	2	0		0	1
	1	1	2		0	1
	2	2	1		2	2
	1	0	2		0	1

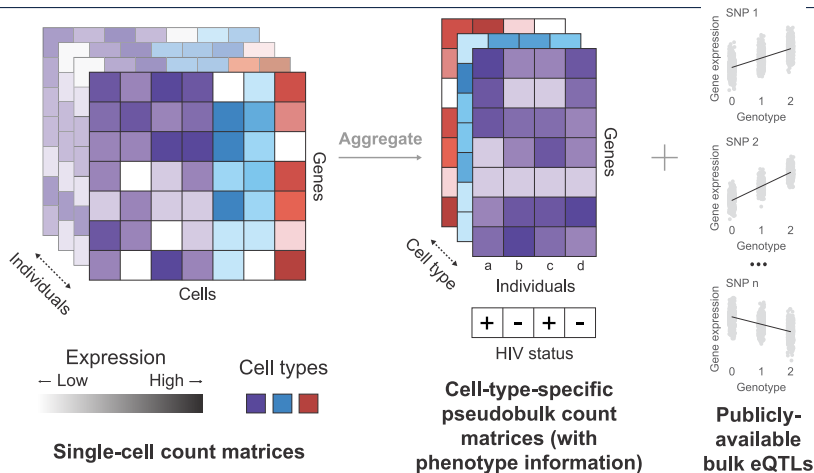
linking



↓
genotype
prediction

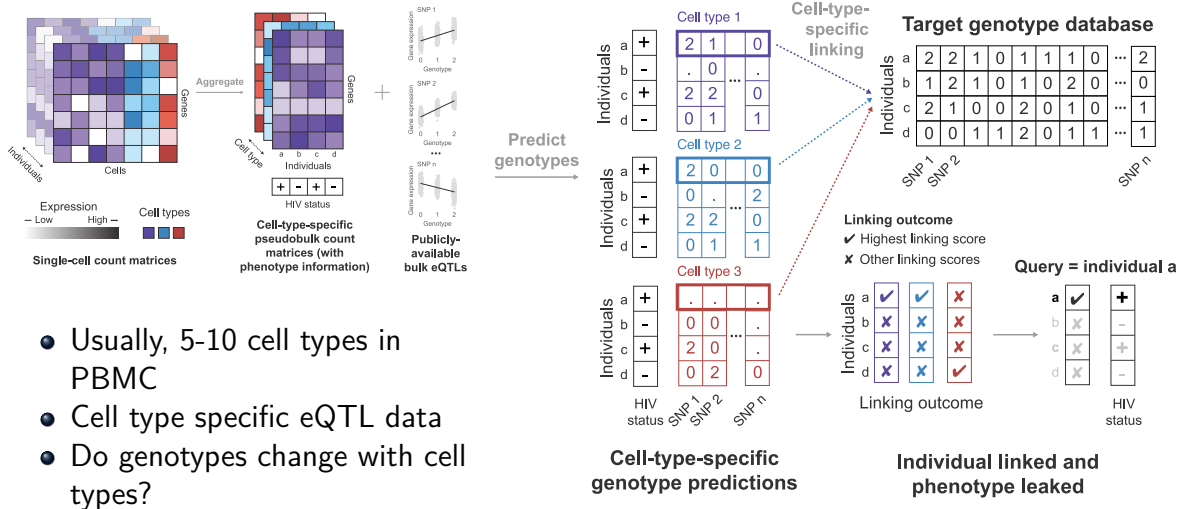
ind	noisy genotypes						pheno
	v_1	v_2	v_3		v_i	v_m	
1	2	2	0		?	2	+
2	2	0	?		0	0	-
3	0	?	2		0	?	-
							+
10	0	2	2		2	2	+
n	0	?	2		?	?	

The way we called eQTLs can create a backdoor

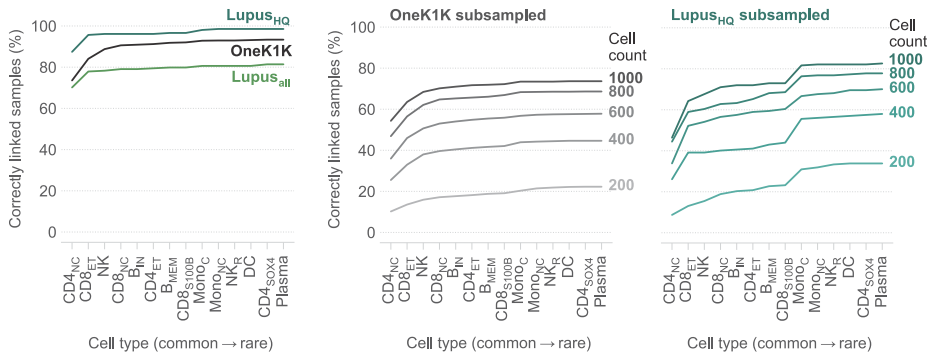


- **Input:** (1) pseudobulk (2) eQTL catalogue

Well, multiple cell types $\approx$ multiple lines of evidence



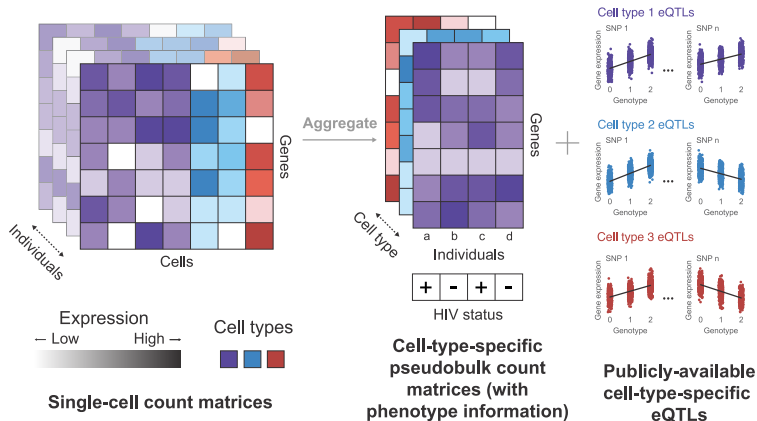
Revealing “donor” information using external DB!



- Rarer cell types $\Rightarrow$ higher linking accuracy (lines trend upward, common $\rightarrow$ rare)
- More cells per type $\Rightarrow$ better pseudobulk estimates $\Rightarrow$ better genotype prediction

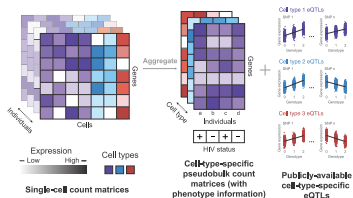
Walker et al. (2024)

What if we have single-cell data and eQTLs?

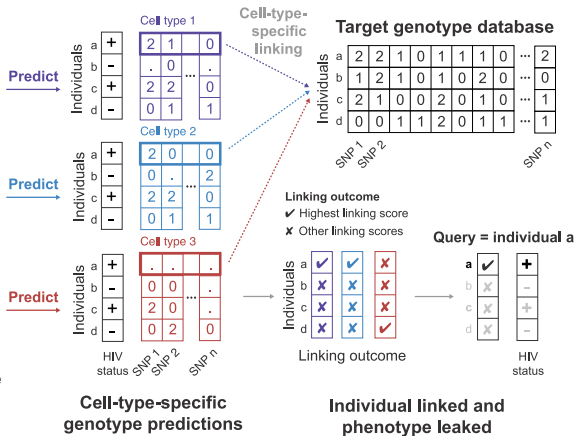


- Can normalization and PCA remove non-genetic individual-specific information?
- Single-cell count matrices aggregated into cell-type-specific pseudobulk

What if we have single-cell data and eQTLs?

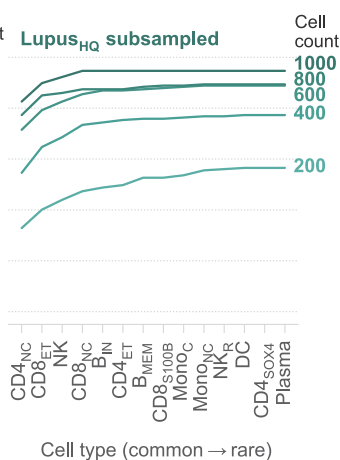
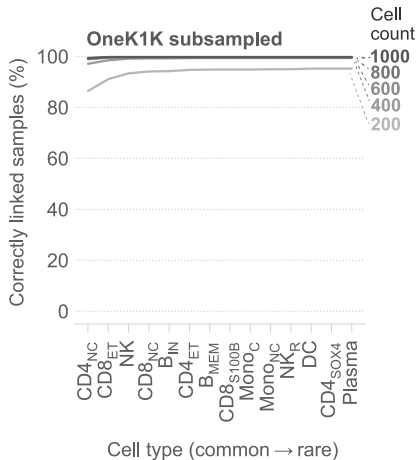
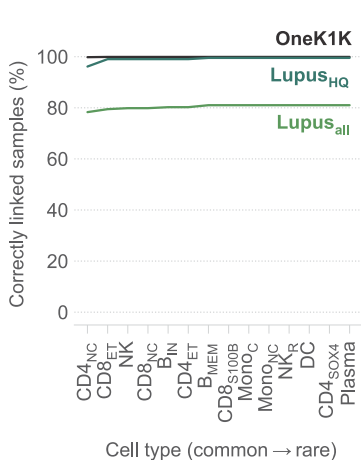


The power of multiple lines of evidence across different cell types



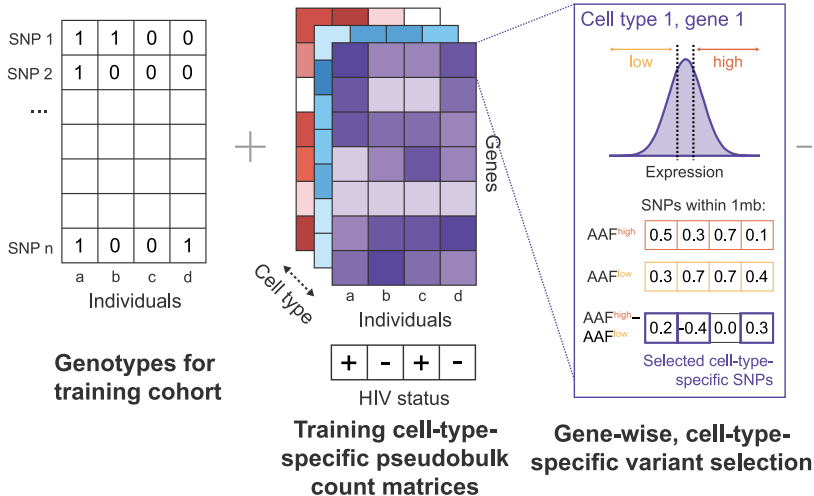
Walker et al. (2024)

Again, strong prediction performance...



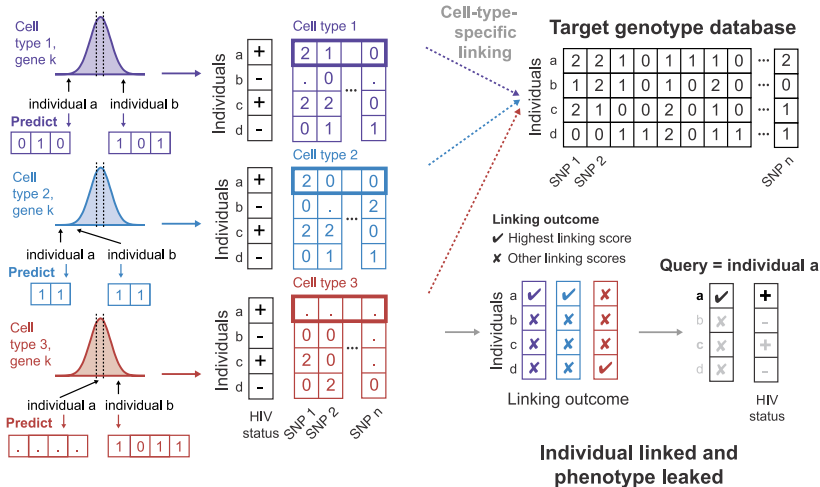
Walker et al. (2024)

Do we even need scQTL summary statistics data?



- No external eQTL catalog needed
- AAAF = alternative allele frequency
- Split by expression (high/low), compare AAAF per locus
- Large $\Delta\text{AAAF} \Rightarrow$ informative variant
- **Cross-cohort:** learn variants from training cohort, attack a different target cohort

Genotypes in training data → predict other cohorts



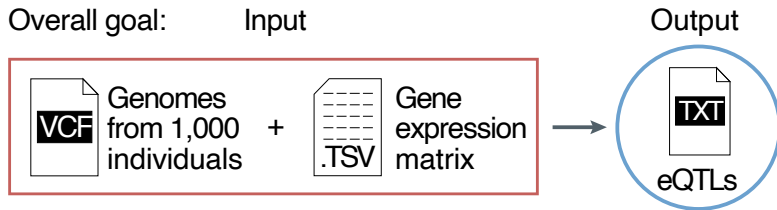
Today's lecture

- 1 Genetic Data Utility vs. Privacy
- 2 Membership inference
- 3 Re-identification attack
- 4 Secured protocols for genomic data analysis**
- 5 Differential Privacy
- 6 Summary

Functional genomics data: privacy risk assessment and technological mitigation

*Gamze Gürsoy , Tianxiao Li , Susanna Liu , Eric Ni ,
Charlotte M. Brannon  and Mark B. Gerstein *

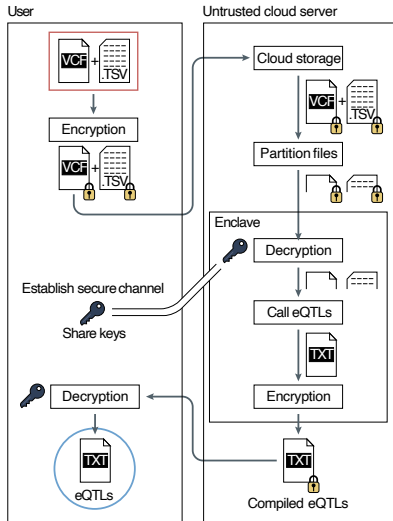
How can we safely estimate eQTLs?



Goal: Protect expression profiled individuals from re-identification attacks

Gürsoy et al. (2022)

Trusted Execution Environments (TEE)



- Data encrypted in transit and at rest
- Decrypted only inside hardware **enclave** (e.g., Intel SGX)
- Even the OS cannot read enclave memory

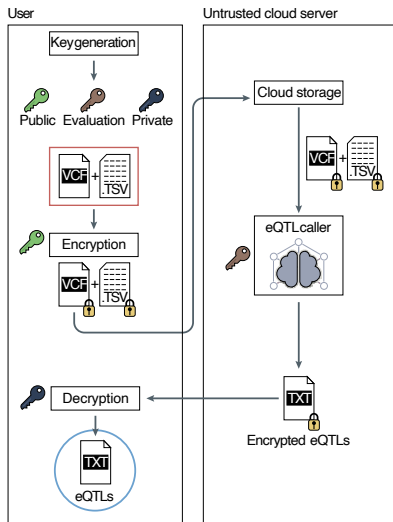
Pros:

- near-plaintext speed
- no utility loss

Cons:

- side-channel attacks
- trust in hardware vendor
- limited enclave memory

Homomorphic Encryption (HE)



Arithmetic on **encrypted data** yields correct results when decrypted:

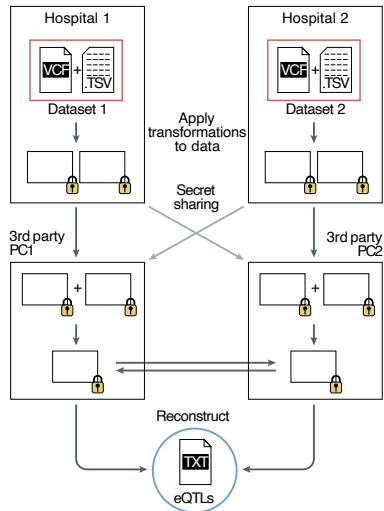
$$\text{Dec}(f(\text{Enc}(x))) = f(x)$$

- Server runs eQTL caller on encrypted genotypes and expression (never sees the actual data)
- Only the user holds the private key to decrypt the result

Pros: strong formal guarantee (computational)

Cons: 10^3 – $10^6 \times$ slowdown vs. plaintext

Secure Multiparty Computation (MPC)

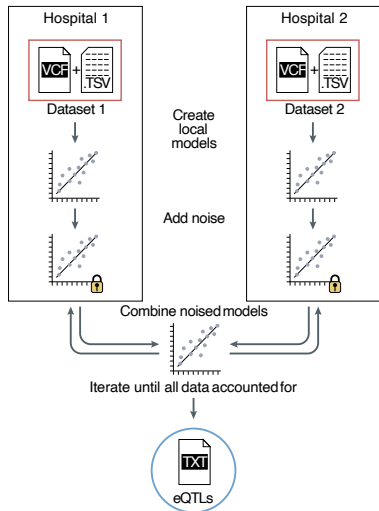


Multiple hospitals pool data **without sharing it**

- Each site splits data into random **secret shares**
- Third-party servers compute on shares
- No single party can reconstruct individual data

Key property: no utility loss. Exact same result as pooled plaintext.

Federated Learning (FL)



Raw data **never leaves** each institution

- 1 Train local models
- 2 Add noise to model updates
- 3 Combine noised models centrally
- 4 Iterate until convergence

Caveat: gradients can leak training data. FL alone provides only informal privacy guarantees.

Today's lecture

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- 6 Summary

LONG LIVE THE REVOLUTION.
OUR NEXT MEETING WILL BE
AT THE DOCKS AT MIDNIGHT
ON JUNE 28 TAB

AHA, FOUND THEM!



WHEN YOU TRAIN PREDICTIVE MODELS
ON INPUT FROM YOUR USERS, IT CAN
LEAK INFORMATION IN UNEXPECTED WAYS.

Differential Privacy: Formal Privacy Guarantees

A randomized mechanism M satisfies ϵ -DP if for all neighboring datasets D, D' :

$$P(M(D) \in S) \leq e^\epsilon \cdot P(M(D') \in S)$$

Intuition: whether or not you participate, published results change by at most e^ϵ .

ϵ	e^ϵ	Interpretation
0.1	1.11	Strong privacy
1	2.72	Moderate
10	$\sim 22,000$	Essentially vacuous

DP for GWAS: Adding Noise to Protect Individuals

Private MAF at a single SNP in n diploid individuals (Uhler et al. 2012; Yamamoto and Shibuya 2021):

$$\hat{f}_{\text{priv}} = \underbrace{\frac{1}{2n} \sum_{i=1}^n x_i}_f + \text{Lap}\left(\frac{1}{n\varepsilon}\right)$$

- Sensitivity: changing one person shifts f by at most $\Delta f = 1/n$
- For $n = 5,000$, $\varepsilon = 1$: noise SD ≈ 0.00028 vs. sampling error ~ 0.01

Tradeoff: DP adds statistical noise. Most painful for **rare variants** where the signal is already small.

DP in Code: Laplace Mechanism

```
# Add Laplace noise for epsilon-differential privacy  
                                # Sensitivity: max change from adding/removing one person  
  
                                # Allele frequency with DP  
  
n <- 5000  
epsilon <- 1.0  
f <- mean(genotypes) / 2 # true MAF  
  
                                # Add Laplace noise: Lap(sensitivity/epsilon)  
  
sensitivity <- 1/n  
noise <- rlaplace(1, scale = sensitivity/epsilon)  
f_private <- f + noise  
  
                                # Publish f_private instead of f
```

Key insight: Controlled noise gives privacy guarantee

DP for Association Tests: Protecting χ^2 Statistics

For a GWAS with genotype $\{0, 1, 2\} \times$ case/control, form a 3×2 contingency table (Uhler et al. 2012):

	Case	Control
$g = 0$	n_{00}	n_{01}
$g = 1$	n_{10}	n_{11}
$g = 2$	n_{20}	n_{21}

Sensitivity (Uhler Thm 3.2): Adding or removing one person changes χ^2 by at most

$$\Delta(\chi^2) = \frac{4N}{N+2}$$

For large N , this is ≈ 4 , while χ^2 for true associations grows with $N \Rightarrow$ relative noise shrinks as sample size grows.

Private χ^2 Release

Private release:

$$\chi_{\text{priv}}^2 = \chi^2 + \text{Lap}\left(\frac{1}{\varepsilon} \cdot \frac{4N}{N+2}\right)$$

Example: $N = 5,000$, $\varepsilon = 1$

- Sensitivity: $(4 \times 5000)/5002 \approx 4.0$
- Noise SD: $4.0/1.0 \times \sqrt{2} \approx 5.7$
- A genome-wide significant SNP has $\chi^2 \approx 30 \Rightarrow \text{signal-to-noise} \approx 5 : 1$

But do we need to release **all** M' test statistics? Noise grows with number released.

Making DP Practical: Use $\log_{10}$ p-Values

Key insight: release $-\log_{10}(p)$ instead of p (Yamamoto and Shibuya 2021).

Sensitivity for a 3×2 table (Yamamoto Prop. 2):

$$\Delta(-\log_{10} p) = \log_{10}(e) \cdot \frac{2N}{N+2}$$

For $N = 100,000$: sensitivity $\approx 0.87 \Rightarrow$ **manageable noise**

Private release:

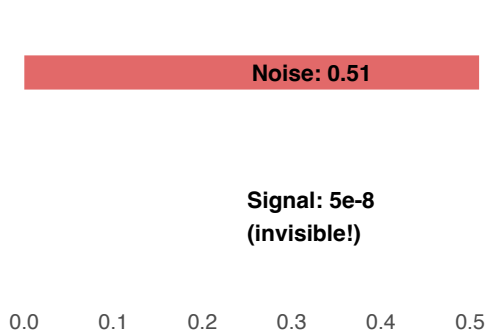
$$(-\log_{10} p)_{\text{priv}} = -\log_{10}(p) + \text{Lap}\left(\frac{0.87}{\epsilon}\right)$$

The significance threshold is now **additive**: $-\log_{10}(5 \times 10^{-8}) = 7.3$

$\Rightarrow$ Compare noisy $-\log_{10}(p)$ directly against 7.3

Raw p-value scale

Signal completely buried by noise



$-\log_{10}(p)$ scale

Signal clearly exceeds noise!



Efficient DP: Release Only Top SNPs, Not All

Uhler Algorithm 1 (Uhler et al. 2012) (3 steps, ϵ -DP by Thm 3.7):

Let $S(\chi^2) = 4N/(N + 2)$. Given M' total SNPs, release top M :

- ➊ **Noisy ranking:** Add $\text{Lap}\left(\frac{4M}{\epsilon} \cdot S(\chi^2)\right)$ to all M' statistics
- ➋ **Select top M :** Pick the M SNPs with largest noisy $\chi^2 \Rightarrow$ set $\mathcal{S}$
- ➌ **Re-noise the winners:** Add *fresh* $\text{Lap}\left(\frac{2M}{\epsilon} \cdot S(\chi^2)\right)$ to the **true** χ^2 values of SNPs in $\mathcal{S}$

Key property: noise scales with M (number released), **not** M' (total SNPs tested).

Safe Z-Score Release via Post-Processing

Post-processing theorem: Any function of a DP output is also DP - no additional privacy cost.

Since $Z^2 = \chi^2$ for 1-df tests, derive a private Z-score from the private χ^2 :

$$Z_{\text{priv}} = \text{sign}(\hat{\beta}) \cdot \sqrt{\max(0, \chi^2_{\text{priv}})}$$

Alternatively, derive from the private $-\log_{10}(p)$:

$$Z_{\text{priv}} = \Phi^{-1}\left(\frac{1}{2} \cdot 10^{-(-\log_{10} p)_{\text{priv}}}\right)$$

This lets downstream consumers use **familiar Z-scores** for meta-analysis, fine-mapping, etc. All for free under DP composition.

Code: Uhler Algorithm 1 + Z-Score Derivation

```
# Uhler Algorithm 1: Release top M SNPs privately
N <- 5000; M <- 100; M_prime <- 500000; eps <- 1.0

                                     # Sensitivity of chi-squared for a 3x2 table
sens_chi2 <- 4 * N / (N + 2)

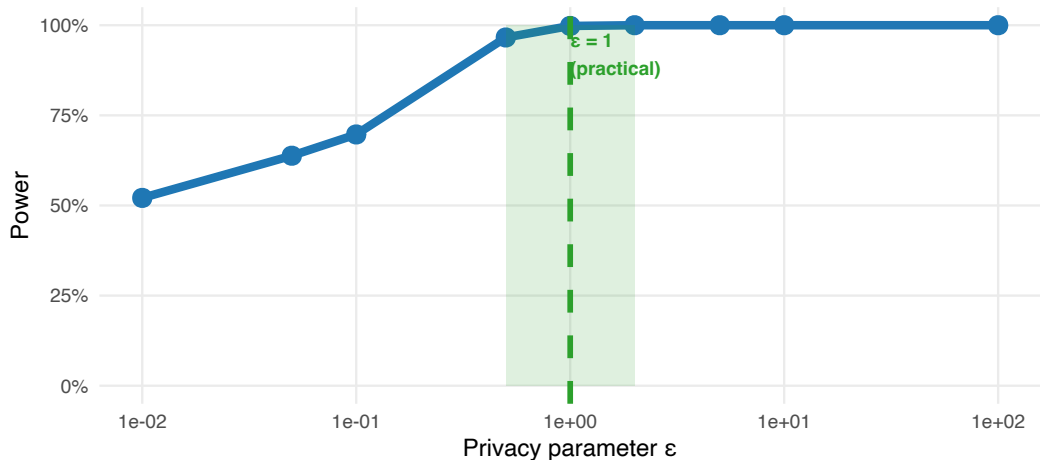
                                     # Step 1: Noisy ranking
chi2_noisy <- chi2_all +
  rlaplace(M_prime, scale = 4 * M / eps * sens_chi2)

                                     # Step 2: Pick top M
top_M <- order(chi2_noisy, decreasing = TRUE)[1:M]

                                     # Step 3: Re-noise the winners (fresh noise on true values)
chi2_private <- chi2_all[top_M] +
  rlaplace(M, scale = 2 * M / eps * sens_chi2)

                                     # Post-process to Z-scores (free by DP composition)
Z_private <- sign(beta_hat[top_M]) *
  sqrt(pmax(0, chi2_private))
```

DP in Action: Privacy vs. Statistical Power



Key insight: DP doesn't mean “no utility”. It means calibrated noise that preserves strong signals while protecting individuals.

Recall: The eQTL Linking Attack

The vulnerability:

- 1 Published expression data for anonymous samples
- 2 Public eQTL catalog (genotype $\leftrightarrow$ expression)
- 3 Reference genotype database

The attack:

Expression $\xrightarrow{\text{eQTL}}$ Predicted genotype $\xrightarrow{\text{match}}$ Individual ID

Result: With 50 genes, individuals can be uniquely identified

Now: Can differential privacy stop this attack?

Can DP Protect Against eQTL Attacks?

Setup: Realistic scenario (50 genes, 500 individuals)

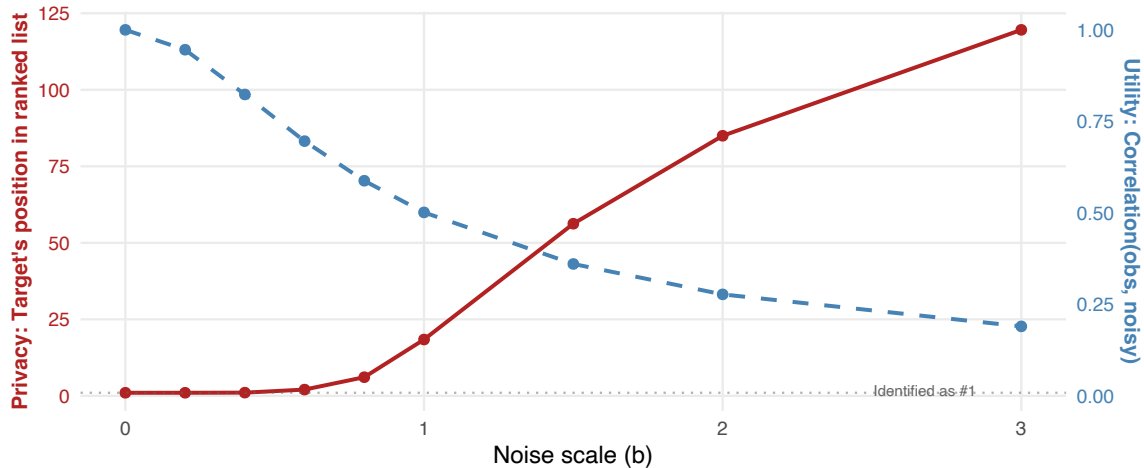
Defense: Add Laplace noise to expression values before publishing

$$e_k^{\text{priv}} = e_k + \text{Lap}(b)$$

Attack: Predict genotypes from noisy expression, then match to database

Question: How much noise is needed to stop the attack?

Does DP Stop the eQTL Attack?



The DP Tradeoff for eQTL Data

What we learned:

Privacy side:

- No noise ($b = 0$): Target at rank ~ 1 (identified)
- Small noise ($b < 0.6$): Rank $\sim 5-10$ (still narrow)
- Moderate noise ($b \geq 1$): Rank $\sim 20+$ (lost in crowd)

Utility side:

- No noise: Perfect correlation ($r=1.0$)
- Moderate noise: Correlation $\sim 0.6-0.7$
- High noise: Data quality severely degraded

Bottom line: DP can protect individuals, but you pay with statistical power for discovery.

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Comparison of approaches

	DP	MPC	HE	FL	TEE
Privacy guarantee	Formal (ϵ)	Info-theoretic	Computational	Informal	Hardware
Utility loss	Yes (noise)	None	None	Some	None
Computational cost	Low	Moderate	High	Moderate	Low
Multi-party native	–	Yes	Limited	Yes	Limited

In practice these are **combined**: e.g., federated learning + DP noise + MPC aggregation.

Genomic privacy

- ❶ **Summary statistics are not safe.** Membership inference from pooled allele frequencies.
- ❷ **Expression data leaks genotypes.** eQTL linking attacks on RNA-seq and scRNA-seq.
- ❸ **Defenses exist but involve tradeoffs.** No single mechanism suffices; layered approaches are the norm.

Nevertheless, information has to be shared

- Why?
- How?
- Ownership?
- Should a government use it?

References

- Erlich, Yaniv, Tal Shor, Itsik Pe'er, and Shai Carmi. 2018. "Identity Inference of Genomic Data Using Long-Range Familial Searches." *Science* 362 (6415): 690–94.
- Gürsoy, Gamze, Tianxiao Li, Susanna Liu, Eric Ni, Charlotte M Brannon, and Mark B Gerstein. 2022. "Functional Genomics Data: Privacy Risk Assessment and Technological Mitigation." *Nat. Rev. Genet.* 23 (4): 245–58.
- Gymrek, Melissa, Amy L McGuire, David Golan, Eran Halperin, and Yaniv Erlich. 2013. "Identifying Personal Genomes by Surname Inference." *Science* 339 (6117): 321–24.

References

- Harmanci, Arif, and Mark Gerstein. 2016. “Quantification of Private Information Leakage from Phenotype-Genotype Data: Linking Attacks.” *Nat. Methods* 13 (3): 251–56.
- Homer, Nils, Szabolcs Szelinger, Margot Redman, et al. 2008. “Resolving Individuals Contributing Trace Amounts of DNA to Highly Complex Mixtures Using High-Density SNP Genotyping Microarrays.” *PLoS Genet.* 4 (8): e1000167.
- Sankararaman, Sriram, Guillaume Obozinski, Michael I Jordan, and Eran Halperin. 2009. “Genomic Privacy and Limits of Individual Detection in a Pool.” *Nat. Genet.* 41 (9): 965–67.

References

- Shringarpure, Suyash S, and Carlos D Bustamante. 2015. “Privacy Risks from Genomic Data-Sharing Beacons.” *Am. J. Hum. Genet.* 97 (5): 631–46.
- Uhler, Caroline, Aleksandra B Slavkovic, and Stephen E Fienberg. 2012. “Privacy-Preserving Data Sharing for Genome-Wide Association Studies.” *arXiv [Stat.ME]*, May.
- Walker, Conor, Arjun Bhatt, Neda Ghaderi, et al. 2024. “Identifying Gene Expression Programs of Cell-Type Identity and Cellular Activity with Single-Cell RNA-Seq.” *Cell*.
- Yamamoto, Akito, and Tetsuo Shibuya. 2021. “More Practical Differentially Private Publication of Key Statistics in GWAS.” *Bioinform. Adv.* 1 (1): vbab004.